

BSs Computer Science
Probability and statistics – 2nd semester - Mid term
March 17, 2026
Duration: 1 hour • Total: 22 points

**Use of phones, documents, or any form of cheating is strictly prohibited.
Any student caught cheating will receive a grade of 0.**

Part I – Short Questions (5 pts)

- **Confidence Interval Interpretation (1 pt)**

Explain precisely what “a 95% confidence interval” means in terms of repeated sampling.

- **Z vs t Interval (1 pt)**

When should we use a Z-interval and when should we use a t-interval?

- **Hypothesis Testing (1 pt)**

What is a p-value?

- **Type I / Type II Errors (1 pt)**

Define Type I and Type II errors. You can use a table for this.

- **Welch-t test (1 pt)**

When the pooled t-test should be used and when the Welch t-test should be used?

Exercise 1 – One-Sample Confidence Interval (3 pts)

A random sample of $n = 36$ students has an average study time of 8.2 hours per week. The population standard deviation is known to be $\sigma = 3$ hours.

- Construct a 95% confidence interval for the true mean study time. (2 pts)
- Interpret this confidence interval. (1 pt)

Exercise 2 – Proportion Inference and Hypothesis Testing (7 pts)

At a university, a survey was conducted to determine which football club students support: Real Madrid or FC Barcelona.

Out of 250 students surveyed, 145 students said they support Real Madrid.

In previous years, surveys suggested that about 50% of students supported Real Madrid.

A Barcelona supporter claims that “Real Madrid fans are now a minority on campus.”

Use statistics to determine whether this claim is supported by the data.

- (A) State the null hypothesis H_0 and the alternative hypothesis H_1 appropriate for testing the Barcelona supporter’s claim. (1 pt)
- (B) Which statistical test should be used in this situation? Explain briefly why this test is appropriate. (1 pt)
- (C) Construct the standard two-sided 95% confidence interval for the population proportion of students who support Real Madrid. (2 pts)
- (D) Construct a 95% lower confidence interval for the proportion of students who support Real Madrid. Interpret. (1 pts)
- (E) Based on your confidence interval, discuss whether the Barcelona supporter’s claim is supported by the data. (1 pt)
- (F) Explain how the width of the confidence interval would change if the sample size were doubled. (1 pt)

Exercise 3 – Hypothesis Test, p-value, and Type II Error (7 pts)

An article in the *Journal of Electronic Material* [“Progress in CdZnTe Substrate Producibility and Critical Drive of IRFPA Yield Originating with CdZnTe Substrates” (1998, Vol. 27, No. 6, pp. 564–572)] improved the quality of CdZnTe substrates used to produce the HgCdTe infrared focal plane arrays (IRFPAs), also defined as sensor chip assemblies (SCAs).

The cut-on wavelength on a sample of 11 wafers was measured and is shown below:

6.06 6.16 6.57 6.67 6.98 6.17 6.17 6.93 6.73 6.87 6.76

- (A) Is there evidence at the 95% confidence level that the mean cut-on wavelength differs from $6.50\ \mu\text{m}$? (2pts)
- (B) Give the approximate p-value for this test using Table C of the appendix. (1 pt)
- (C) Compute the 95% confidence interval for the population mean cut-on wavelength ($6.50\ \mu\text{m}$). (2pt)
- (D) What is the Type II error probability if the true mean cut-on wavelength is $6.95\ \mu\text{m}$? (2 pt)

Hint: When computing the Type II error probability, use the normal (Z) distribution, because we evaluate the probability of the sample mean under the true mean $\mu = 6.95\ \mu\text{m}$.

If $z < -3.0$, then $P(Z < z) \approx 0$.

If $z \approx -2$, then $P(Z < z) \approx 0.02$.

Appendix – Statistical Tables

Table A. Standard Normal Critical Values for Two-Tailed Tests and one tailed test

Values shown are the critical values $z_{\alpha/2}$ for a **two-sided test** with significance level α .
Use this table for **confidence intervals** or **two-sided hypothesis tests**.

Total significance level α	Area in each tail $\alpha/2$	Critical value $z_{\alpha/2}$
0.10	0.05	1.645
0.05	0.025	1.96
0.02	0.01	2.326
0.01	0.005	2.576

Values shown are the critical values z_{α} for a **one-sided test** with significance level α .

Significance level α	Critical value z_{α}
0.10	1.28
0.05	1.645
0.025	1.96
0.01	2.326
0.005	2.576

Table B. One tail and two tails t-Table

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850

Table C. Two-tailed p-values for the t-distribution

Values shown are $p = P(|T_{df}| > t)$. Use the column that matches the closest degree of freedom.

t	df = 5	df = 8	df = 10	df = 15	df = 20
0.0	1.000	1.000	1.000	1.000	1.000
0.2	0.848	0.846	0.845	0.844	0.844
0.3	0.782	0.780	0.779	0.778	0.778
0.4	0.710	0.707	0.706	0.705	0.705
0.5	0.638	0.636	0.635	0.634	0.633
0.7	0.520	0.516	0.514	0.512	0.511
1.0	0.363	0.347	0.340	0.332	0.327
1.3	0.248	0.230	0.223	0.214	0.209
1.6	0.167	0.150	0.142	0.134	0.130
2.0	0.101	0.077	0.071	0.064	0.060
2.3	0.067	0.045	0.039	0.033	0.030
2.6	0.046	0.028	0.023	0.019	0.017
3.0	0.030	0.017	0.013	0.009	0.007